

Lab: MongoDB Indexing



Objectives

After completing this lab you will be able to:

- Measure the time it takes to execute a query with the explain function
- Describe the process of creating, listing and deleting indexes
- Evaluate the effectiveness of an index

Exercise 1 - Get the environment ready

Open the MongoDB database page by clicking the button below:

 Open MongoDB Page in IDE

On that page, click the **Create** button to create a MongoDB database.

After creation has finished, click the **MongoDB CLI** button (below the **Create** button from the previous step) to connect to the database using the mongosh CLI.

1. Problem:

Select the training database.

▼ Click here for Hint

Refer to the [MongoDB Getting Started lab](#).

▼ Click here for Solution

```
1 use training
```

2. Problem:

Create a collection named bigdata.

▼ Click here for Hint

Refer to the [MongoDB Getting Started lab](#).

▼ Click here for Solution

```
1 db.createCollection("bigdata")
```

Exercise 2 - Insert documents

- Let us insert a lot of documents into the newly created collection.
- This should take around 3 minutes, so please be patient.
- The code given below will insert 200000 documents into the 'bigdata' collection.
- Each document would have a field named `account_no` which is a simple auto increment number.
- And a field named `balance` which is a randomly generated number, to simulate the bank balance for the account.

Copy the below code and paste it on the mongo client.

```
1 use training
2 for (i=1;i<=200000;i++){print(i);db.bigdata.insert({"account_no":i,"balance":Math.round(Math.random()*1000000)}})
```

Verify that 200000 documents got inserted by running the below command.

```
1 db.bigdata.countDocuments()
```

The screenshot shows the MongoDB Atlas interface. On the left, a sidebar contains navigation icons and a menu with options: DATABASES, BIG DATA, CLOUD, EMBEDDABLE AI, and OTHER. Below these is a 'Launch Applica...' button. The main panel has a 'Create' button and a 'Delete' button. Below these are tabs for 'Summary', 'Connection Information', and 'Details'. The 'Summary' tab is active, displaying a message: 'Your database and mongo-express server are now ready to use and available with the following login credentials. For more details on how to navigate MongoDB, please check out the Details section.' Below this message, there are two buttons: 'mongo-express' and 'MongoDB CLI'. The 'MongoDB CLI' button is highlighted. Below the buttons, there is a terminal window showing the command prompt 'training>' and the output of the 'db.bigdata.countDocuments()' command, which is '200000'. The terminal also shows the command 'db.bigdata.insert({"account_no":i,"balance":Math.round(Math.random()*1000000)})' being executed.

Exercise 3 - Measure the time taken by a query

Let us run a query and find out how much time it takes to complete.

Let us query for the details of account number 58982.

We will make use of the explain function to find the time taken to run the query in milliseconds.

Run the below command.

```
1 db.bigdata.find({"account_no":58982}).explain("executionStats").executionStats.executionTimeMillis
```

Make a note of the milliseconds it took to run the query. We will need this at a later stage.

```
training> db.bigdata.find({"account_no":58982}).explain("executionStats").executionStats.executionTimeMillis
72
training> █
```

Exercise 5 - Find out how effective an index is

Let us run a query and find out how much time it takes to complete, using an index.

Let us query for the details of account number 69271.

Run the below command.

```
1 db.bigdata.find({"account_no": 69271}).explain("executionStats").executionStats.executionTimeMillis
```

Make a note of the milliseconds it took to run the query. Compare this with the time you recorded during Exercise-3. This should be a lot less than that.

```
training>
... db.bigdata.createIndex({"account_no":1})
account_no_1
training> db.bigdata.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { account_no: 1 }, name: 'account_no_1' }
]
training> db.bigdata.find({"account_no": 69271}).explain("executionStats").executionStats.executionTimeMillis
1
training> █
```

Exercise 6 - Delete an index

Use the below command to delete the index we created earlier.

```
1 db.bigdata.dropIndex({"account_no":1})
```